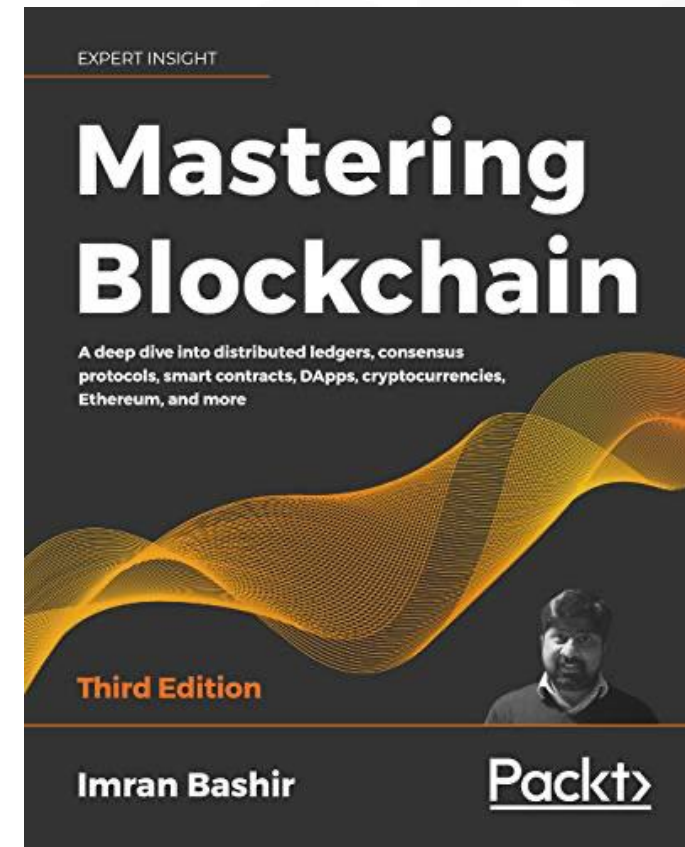


Mastering Blockchain

Third Edition

Chapter 21, Scalability and Other
Challenges



Outline

- Scalability
- Privacy
- Security
- Other challenges



Scalability – a layered approach

We can divide approaches to solving scalability issues into three main categories based on the layer in the blockchain stack they operate on:

- *Layer 0* methods, or *network* layer methods, where the mechanisms to improve scalability operate at the network level of the blockchain architecture stack
- *Layer 1* methods are also called on-chain methods, where the blockchain protocol itself is enhanced to improve scalability. This layer represents the blockchain itself
- *Layer 2* methods, or off-chain methods, where mechanisms that exist outside of the main blockchain are used to improve the scalability of the blockchain

Scalability solutions



Scalability can be achieved at different layers:

- Layer 0 methods include Kadcast, bloXroute
- Layer 1 methods include transaction parallelization, increased block size
- Layer 2 methods include state channels

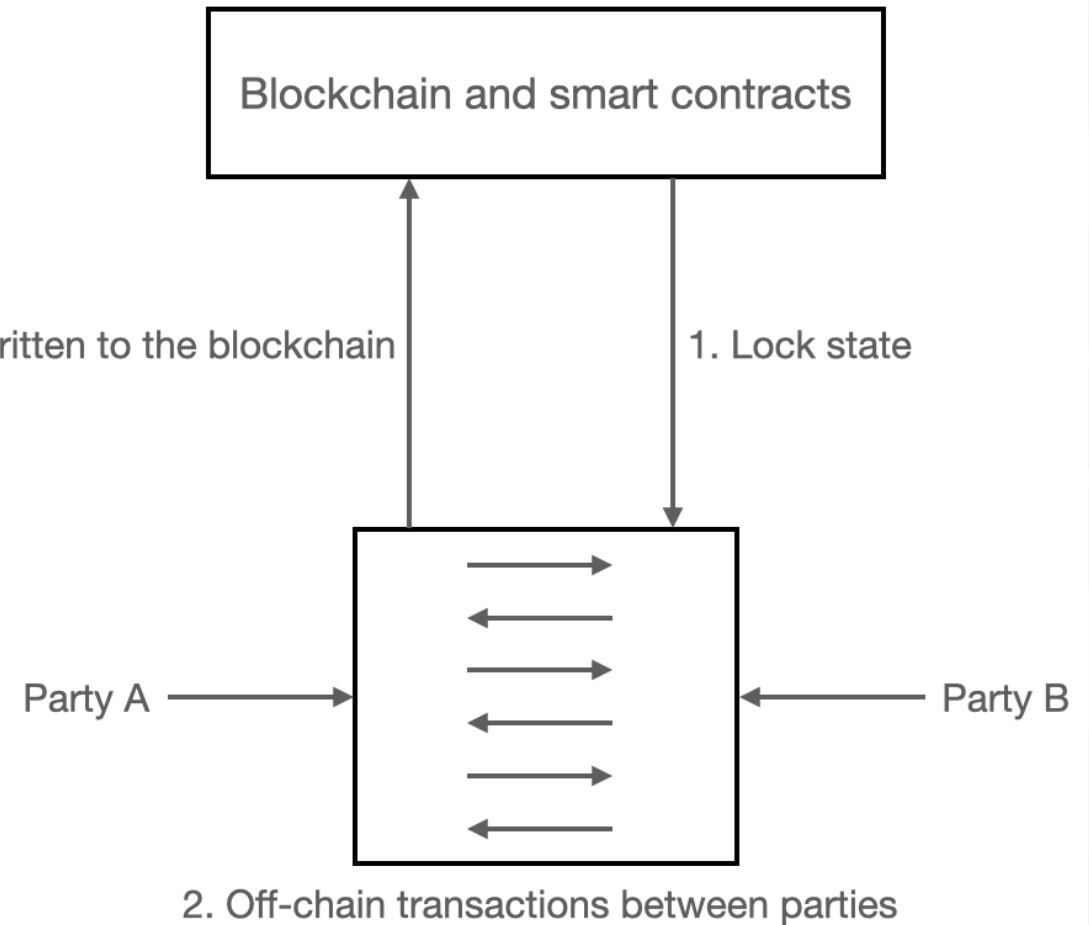
State channels

The key idea of state channels is to use side channels for state updating and processing transactions off the main chain.

Once the state is finalized, it is written back to the main chain, thus offloading the time consuming operations from the main blockchain.

It is typically a three step process :

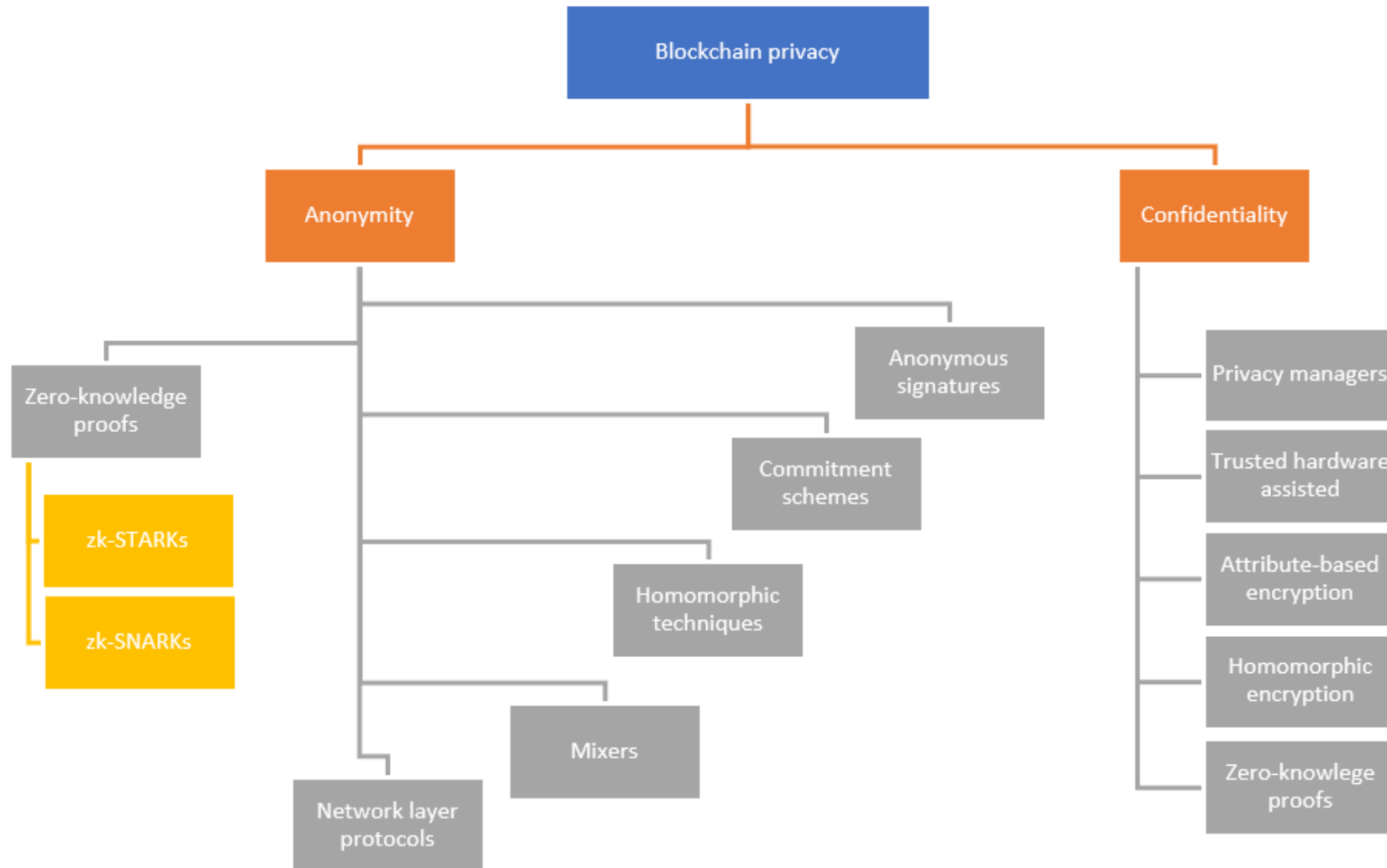
1. A part of the blockchain state is locked under a smart contract
2. Off-chain transaction occurs
3. Once the final state is achieved, the state channel is closed, and the final state is written back to the main blockchain. At this stage, the locked part of the blockchain is also unlocked



Privacy solutions

- Layer 0
 - Tor
 - I2P
- Layers 1 and 2 (as many of the techniques can be used at Layers 1 and 2—for example, zero-knowledge proofs—we are not distinguishing between these layers)
 - Indistinguishability obfuscation
 - Zero-knowledge proofs
 - Bulletproofs
 - Mixing protocols
 - Confidential transactions
 - MimbleWimble
 - Ring signatures

High-level taxonomy of blockchain privacy techniques



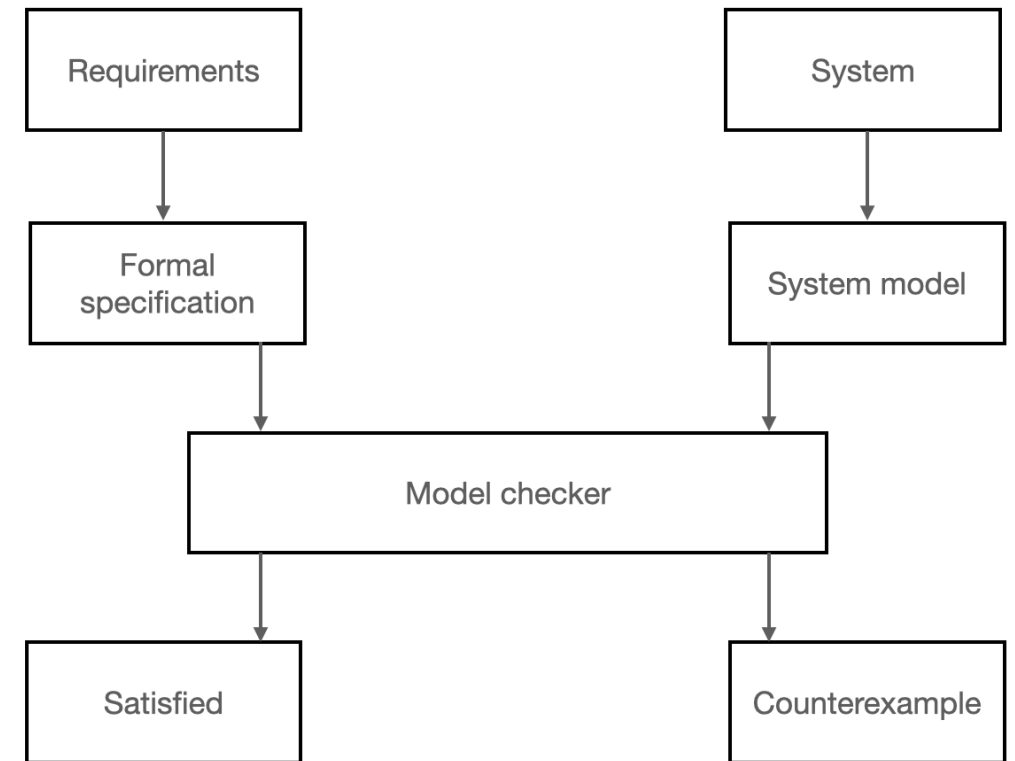
Security solutions

8

Formal verification methods are the set of techniques used to model systems as mathematical objects.

Model checking is a formal verification technique used to verify finite state systems automatically, with an exhaustive search of the state space:

1. The model checking process starts with modelling, where a formal specification is created from an informal design of the model
2. The specification step comes next, where relevant properties of the design are specified using some logical formalism
3. Finally, verification of the model is carried out, which checks all the properties defined in the specification and provides results. In case of negative outcomes, error traces with counterexamples are provided to help the designer trace the error



Smart contract security



- Common bugs
 - Reentrancy
 - Timestamp dependence
 - Randomness
 - Integer overflow
 - Integer underflow
- Analysis
 - Static analysis
 - Oyente
 - Why3

Other challenges

- Interoperability/pluggability
- Lack of standardization
- Post-quantum resistance
- Compliance and regulation

Exercise



- Using Remix IDE's static analysis capability, analyze a smart contract for vulnerabilities.

Summary

In this presentation, we explored:

- Challenges facing blockchain technology
- Layered approach to solving blockchain challenges
- Scalability and state channels
- Privacy and Tor, I2P, mixers, zero-knowledge proofs
- Security and formal verification
- Smart contract security
- Other challenges to consider, such as standardization and quantum resistance.